

Agentic AI for Pattern Recognition and Decision Making in Healthcare Payment Integrity

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Introduction and Topic Definition

Healthcare fraud, waste, and abuse (FWA) is one of the most expensive problems in the U.S. system. In 2023 the Centers for Medicare and Medicaid Services reported \$14.9 billion in FWA recoveries (ICF, 2026), and yearly losses are estimated in the tens to hundreds of billions of dollars (Elevance Health, 2026). For a payment integrity company like Cotiviti, telling improper claims apart from legitimate ones is the core of the business.

This report looks at *agentic generative AI* for clinical decision making and pattern recognition across Treatment, Payment, and Operations (TPO), with a focus on the Payment side. A normal language model answers a single prompt. An agentic system sets a goal, reasons through it, and takes actions to reach it (Thummarakoti, 2026), usually by calling external tools and combining the results into a decision. For claims, this means an agent can gather evidence about a flagged claim on its own, check it against coding and payment rules, and recommend what to do with it.

The hard part is not detection. It is *precision*. Statistical models flag many suspicious claims, but most of those alerts are false positives, which wastes investigation time and burns out analysts (Fraud.net, 2025). Because of this, the field is moving away from reactive recovery and toward layered systems that pair machine learning detection with investigation that reasons about each case (Masood, 2025). This report studies that shift, presents a working prototype I built to show it, and proposes options for Cotiviti.

Relevant Trends

Payment integrity technology has moved through three stages: fixed billing rules, then statistical and machine learning anomaly detection (Nabrawi and Alanazi, 2023), and now agentic systems that flag suspicious claims and also investigate them on their own. Practitioners describe agentic AI as a way to connect siloed payment integrity teams and move information between them more easily (SAS, 2025). These agents do not replace detection models. They add a layer of reasoning on top of them.

A key part of this shift is keeping a human in the loop. Recent designs let agents handle routine, low risk claims automatically and send only the unclear or high impact cases to a human reviewer (Thummarakoti, 2026). The business case is also getting clearer. Insurers using AI across claims intake, fraud scoring, and adjudication report 20 to 35 percent lower operating costs and up to 50 percent faster claims cycles (Mobisoft, 2026).

What now limits adoption is trust, not raw model ability. When I built an agentic system with the Model Context Protocol, the hard part was not connecting tools. It was getting the agent to reason correctly and consistently. This is why reliability, governance, and explainability have become the main concerns in this area rather than novelty (ICF, 2026).

Opportunities and Threats

Opportunities. The biggest opportunity is precision. In my own work building anomaly detection models on messy transactional data, the main cost was not missed fraud. It was the volume of false positives, which used up analyst time and made people trust the system less. An agent that investigates each flagged claim against policy and provider history before it reaches a person can remove much of that noise, which detection alone cannot do. Teams that apply tailored machine learning to this problem report false positive reductions of up to 90 percent (Fraud.net, 2025). Other benefits include faster review, fully auditable decisions, and the ability to handle more claims without hiring at the same pace, while reusing the detection models a payer already owns.

Threats. The risks are real. Large language models can make things up or reason inconsistently, the same difficulty I ran into, so reliability is the first concern for any claims decision. There is also real regulatory and compliance exposure, along with the handling of protected health information. The literature stresses that governance has to be built into the data and model lifecycle from the start, with explanations that give clear reasons for each flag (ICF, 2026). Peer reviewed work shows that explainable, unsupervised detection workflows can help human investigators in real insurance settings (De Meulemeester et al., 2025). A quieter risk is over automation. Taking human judgment out of serious decisions invites both mistakes and a loss of accountability. Model drift and vendor lock in round out what a payer would need to watch.

Strategic Recommendations for Cotiviti

Cotiviti is in a strong position to lead this change because it already runs mature detection systems. The opportunity is to add reasoning and governance on top of what exists, not to rebuild. I recommend a staged approach that starts small.

First, start with a bounded pilot. Instead of rolling agentic triage out across every claim type, Cotiviti should pick one high volume, well understood claim category and add an agentic investigation layer there. A small pilot makes the agent's reasoning easier to check, keeps risk contained, and produces real evidence such as lower false positive rates and analyst time saved before any wider rollout. The lesson from practice is simple: agent reliability has to be proven on a narrow problem before it can be trusted at scale (Thummarakoti, 2026).

Second, treat governance as an advantage, not a burden. In a regulated market, an agent that logs its work and can justify every recommendation is a selling point (ICF, 2026). Building audit trails and human review into the pilot from day one puts Cotiviti ahead of competitors who add these later. The prototype I built shows this is feasible. On a synthetic set of 355 claims seeded with five common FWA patterns, an unsupervised Isolation Forest, a method already well established for healthcare insurance fraud detection (Nabrawi and Alanazi, 2023; Moradi et al., 2025), flagged the suspicious claims and caught 62 percent of the seeded ones, while 37 percent of its flags were false alarms, which is exactly the noise problem described above. An agent then investigated each flagged claim with tools that check code compatibility, units, duplicates, and provider history, returned a decision with a plain language reason and a confidence score, and passed it through a governance step that sends serious cases to a person and logs everything. The takeaway is simple. **Cotiviti's fastest path to value is not the biggest deployment but the most carefully chosen one: prove agentic payment integrity on a narrow front, with governance built in, then scale what works.**

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